HERPES SIMPLEX

AT A GLANCE

Herpes simplex viruses (HSVs) are common human DNA viral pathogens that intermittently re-activate. After replication in the skin or mucosa, the virus infects local nerve endings and ascends to the ganglia, where it establishes latency until re-activation.

There are two types of HSV: HSV-1 and HSV-2. HSV-1 is primarily associated with orofacial infections, while HSV-2 typically causes genital infections. However, both types can infect oral and genital areas and cause acute and recurrent disease.

Most adults are seropositive for HSV-1, with the majority of infections acquired during childhood. Approximately one-fourth of adults in the United States are infected with HSV-2, with acquisition closely linked to sexual activity.

Most primary HSV infections are asymptomatic or unrecognized, though they can occasionally cause severe disease. Similarly, most recurrences are asymptomatic—referred to as asymptomatic shedding—and this is the primary mode of transmission.

Genital herpes is the most prevalent sexually transmitted infection worldwide and the leading cause of ulcerative genital disease. It is also a significant risk factor for the acquisition and transmission of human immunodeficiency virus (HIV).

HSV can also cause infections of the eye (e.g., keratitis), central nervous system (e.g., encephalitis), and neonatal herpes. Defects in cellular immunity increase the risk of severe, disseminated, or chronic HSV infections.

Diagnosis is based on viral culture, polymerase chain reaction (PCR), and serology, depending on the clinical context.

Treatment involves antiviral agents such as **acyclovir**, **valacyclovir**, or **famciclovir**. Regimens and dosages vary by clinical presentation. Antiviral resistance is rare except in immunocompromised individuals.

After DNA replication, the expression of more than 60 "late" HSV-2 genes occurs. These genes encode virion structural components and proteins required for proper viral assembly.

PATHOGENESIS

In vivo, HSV infection progresses through three stages: acute infection, establishment and maintenance of latency, and re-activation.

During acute infection, the virus replicates at the site of inoculation on mucocutaneous surfaces, producing primary lesions. Virus particles then spread to sensory nerve terminals and undergo retrograde axonal transport to neuronal nuclei in regional sensory ganglia.

In a subset of infected neurons, a latent infection is established. During latency, viral DNA persists as an episome, and gene expression is highly restricted—only one viral transcript, the latency-associated transcript (LAT), is abundantly expressed.

In the final stage, re-activation occurs, leading to viral replication and anterograde axonal transport of newly assembled virions to peripheral sites, often near the original site of entry.

HSV-1 reactivates most frequently from the trigeminal ganglia, whereas HSV-2 primarily reactivates from sacral ganglia. The rate of reactivation is influenced by the quantity of latent viral DNA in the ganglia. Additionally, type-specific genomic sequences within HSV contribute to differences in recurrence rates at specific anatomical sites.

Host factors also play a critical role in reactivation. Experimental models show that reactivation can be triggered by ultraviolet irradiation, hyperthermia, local trauma, and other physiological stressors—conditions that similarly induce reactivation in humans.

IMMUNE RESPONSE

Host immunity significantly affects susceptibility to HSV infection, disease severity, and recurrence frequency. The level of cellular immune competence strongly correlates with clinical outcomes.

Individuals with mild impairment in cellular immunity may experience more frequent recurrences and delayed healing of lesions. In contrast, those with severe immunosuppression are at higher risk for disseminated, chronic, or drug-resistant infections.